

Propagation of high-energy particles: Electrons

(Lecture 9)

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Outline of this chapter

- Introduction: Difference from nuclei.
- Galactic propagation of cosmic-ray electrons
 - Spectrum from far away sources
 - Spectrum from nearby sources

Introduction

- Propagation of CR electrons differs from that of the nuclei.
- Both particle can suffer ionization, coulomb and adiabatic losses.
- But, electrons can suffer additional losses due to Bremsstrahlung, synchrotron and inverse-Compton processes.

Galactic propagation of CR electrons

The transport equation for electrons

As in the case of nuclei, the transport equation can be written as a diffusion-loss equation:

$$\frac{dN(E)}{dt} = D \nabla^2 N(E) + \frac{\partial}{\partial E} [b(E)N(E)] + Q(E) . \quad (16.25)$$

This expression describes the evolution of the energy spectrum as the particles diffuse from their sources under conditions of continuous energy losses described by $-(dE/dt) = b(E)$.

The form of $b(E)$ for high energy electrons under interstellar conditions is the sum of all the processes discussed in Sect. 16.2, namely,

$$b(E) = -\left(\frac{dE}{dt}\right) = A_1 \left(\ln \frac{E}{m_e c^2} + 19.8 \right) + A_2 E + A_3 E^2 . \quad (16.28)$$



Ionization losses Bremsstrahlung Inverse Compton
+ Synchrotron

Galactic propagation of CR electrons

Some simple cases first before solving the full transport equation:

A steady state case without diffusion:

Rather than solve the full diffusion-loss equation, let us simplify the problem by neglecting diffusion and assume that the spectrum has reached a steady state under the continuous injection of particles described by $Q(E)$ and energy losses given by $b(E)$. First of all, we assume there is an infinite, uniform distribution of sources, each injecting high energy electrons with an injection spectrum $Q(E) = \kappa E^{-p}$. Then, all spatial dependence disappears from the solution and the diffusion-loss equation reduces to

$$\frac{d}{dE}[b(E) N(E)] = -Q(E) \quad \text{or} \quad \int d[b(E) N(E)] = - \int Q(E) dE . \quad (16.26)$$

Galactic propagation of CR electrons

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Assuming $N(E) \rightarrow 0$ as $E \rightarrow \infty$, (16.26) can be integrated to give

$$N(E) = \frac{\kappa E^{-(p-1)}}{(p-1)b(E)} . \quad (16.27)$$

The electron spectrum will be shaped by the effect of the energy loss process $b(E)$.

Galactic propagation of CR electrons

A steady state case without diffusion:

We have from the previous slide: $N(E) = \frac{\kappa E^{-(p-1)}}{(p-1)b(E)}$. (16.27)

Recall: $b(E) = -\left(\frac{dE}{dt}\right) = A_1 \left(\ln \frac{E}{m_e c^2} + 19.8\right) + A_2 E + A_3 E^2$. (16.28)



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↑
Inverse Compton
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Galactic propagation of CR electrons

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- (ii) if bremsstrahlung or adiabatic losses dominate, $N(E) \propto E^{-p}$ – the spectrum is unchanged; For $100 < E < 10$ GeV

Galactic propagation of CR electrons

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- (ii) if bremsstrahlung or adiabatic losses dominate, $N(E) \propto E^{-p}$ – the spectrum is unchanged; For $100 < E < 10$ GeV
- (iii) if inverse Compton or synchrotron losses dominate, $N(E) \propto E^{-(p+1)}$ – the spectrum is steeper by one power of E . For $E > 10$ GeV

Galactic propagation of CR electrons

Time-dependent case neglecting diffusion:

(We assume the energy loss is dominated by inverse Compton and synchrotron processes)

The transport equation is: $\frac{dN(E)}{dt} = \frac{\partial}{\partial E}[b(E)N(E)] + Q(E)$ where, $b(E) = a E^2$

Galactic propagation of CR electrons

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Case I (continuous injection of particles): $Q(E) = \kappa E^{-p}$ at all times.

The solution is:

$$N(E) = \begin{cases} \frac{\kappa E^{-(p+1)}}{a(p-1)} [1 - (1 - aEt)^{p-1}] & \text{if } E \leq E_0 \\ \frac{\kappa E^{-(p+1)}}{a(p-1)} & \text{if } E > E_0 \end{cases} , \quad (16.29)$$

$$\text{if } E > E_0 \quad . \quad (16.30)$$

where $E_0 = 1/(at)$ is the energy at which the energy loss time of the electron becomes equal to time t .

Galactic propagation of CR electrons

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where $E_0 = 1/(at)$ is the energy at which the energy loss time of the electron becomes equal to time t .

This shows that for $E \leq E_0$, the electron spectrum still reflects the source spectrum: $N(E) \propto E^{-p}$
For $E > E_0$, the spectrum is steeper due to the fast energy loss rate: $N(E) \propto E^{-(p+1)}$

Galactic propagation of CR electrons

Time-dependent case neglecting diffusion

Case II (burst-like injection of particles at time $t=0$):

The transport equation is again: $\frac{dN(E)}{dt} = \frac{\partial}{\partial E}[b(E)N(E)] + Q(E)$ where, $b(E) = a E^2$

The source term becomes: $Q(E) = \kappa E^{-p} \delta(t)$,

Galactic propagation of CR electrons

Time-dependent case neglecting diffusion

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The solution in this case is: $N(E) = \kappa E^{-p} (1 - aEt)^{p-2}$.

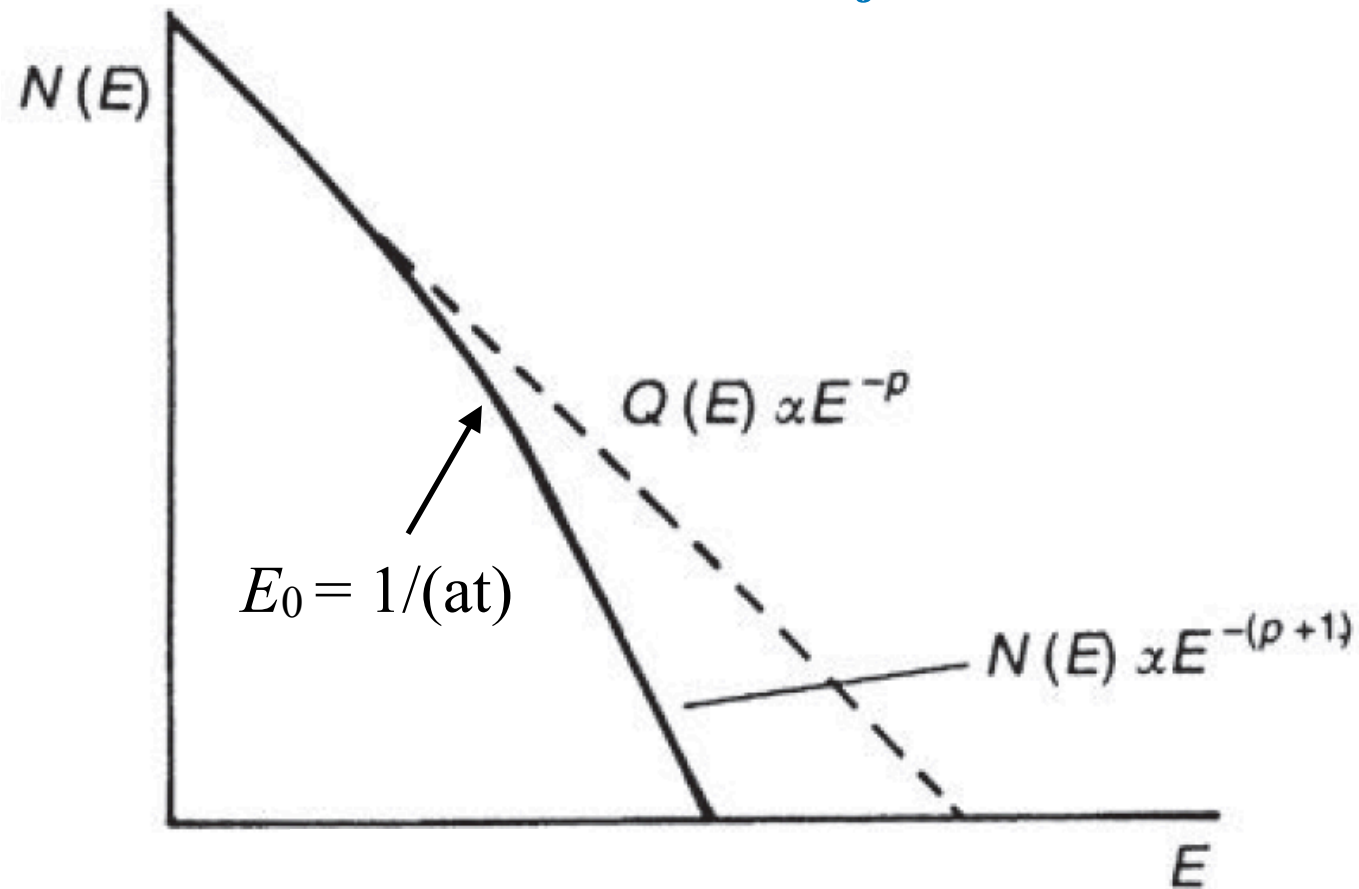
At $E \leq E_0$, the electron spectrum follows the source spectrum: $N(E) \propto E^{-p}$

Then, the spectrum then smoothly cuts off at $E_0 = 1/(at)$. So, at $E > E_0$, $N(E) = 0$.

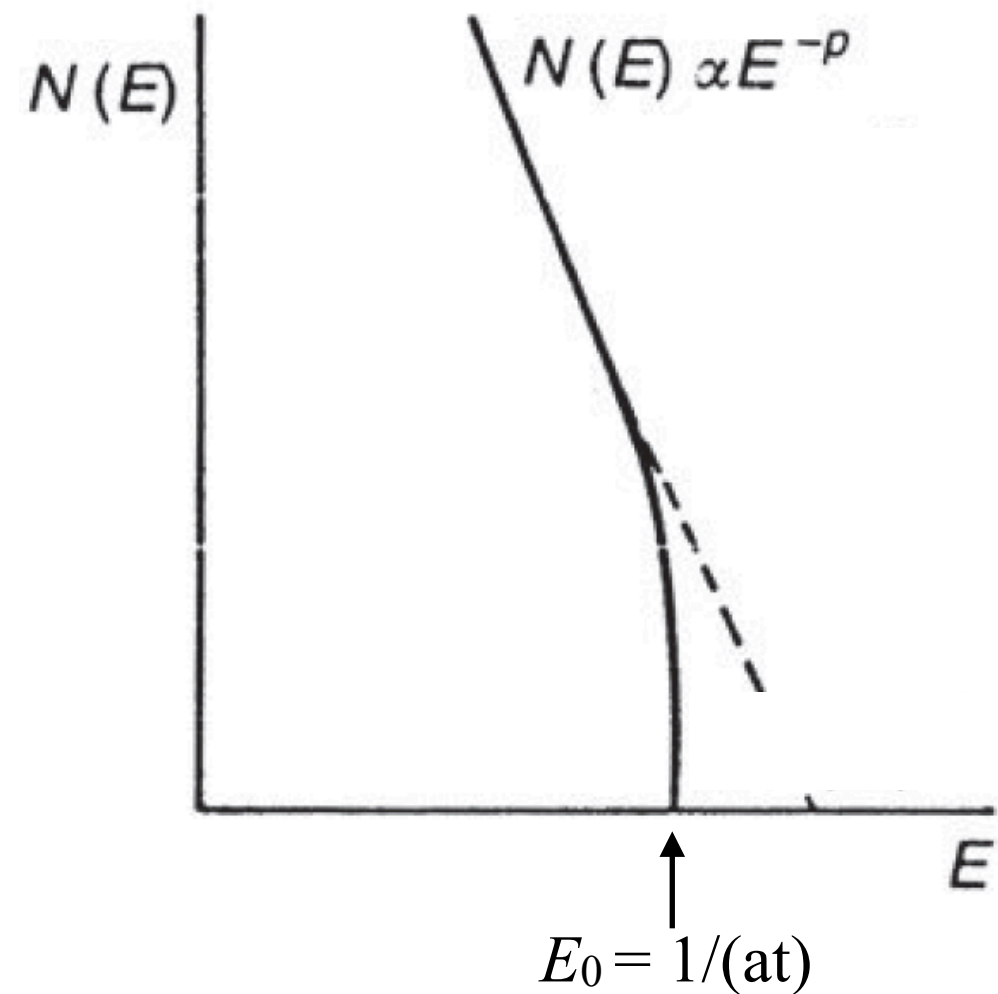
Galactic propagation of CR electrons

Time-dependent case neglecting diffusion

Case I: Continuous injection



Case II: Burst-like injection at $t=0$



Galactic propagation of CR electrons

The full transport equation in the Galaxy including diffusion and energy loss

Steady state equation: $\nabla \cdot (D \nabla n_e) + \frac{\partial}{\partial E} \{b(E) n_e\} = -Q_e$ (8)

where $n_e(\mathbf{r}, E)$ is the electron number density in the Galaxy, $D(E)$ is the diffusion coefficient which is assumed to be constant through out the Galaxy and $Q_e(\mathbf{r}, E)$ is the source term given by $Q_e(\mathbf{r}, E) = \mathfrak{R} q(E) \delta(z)$ assumed to be in the Galactic disk. $\mathfrak{R}$ denotes the rate of supernova explosion per unit surface area on the Galactic disk.

Note: This steady state equation applies to a source distribution in the Galaxy which are continuously releasing cosmic rays (stationary sources).

Galactic propagation of CR electrons

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$\Re$ denotes the rate of supernova explosion per unit surface area on the Galactic disk.

$$n_e(E) = \frac{\Re}{2\sqrt{\pi}b(E)} \int_E^\infty dE' q(E') \left[\int_E^{E'} \frac{D(u)}{b(u)} du \right]^{-1/2} \quad (9)$$

For the diffusion coefficient $D(E) \propto E^\delta$, Eq. (9) shows that

$$n_e(E) \propto \frac{1}{b(E)} \int_E^\infty dE' \frac{q(E')}{\sqrt{E^{\delta-1} - E'^{\delta-1}}} \quad \text{where, we take } b(E) = a E^2.$$

Galactic propagation of CR electrons

The full transport equation in the Galaxy including diffusion and energy loss

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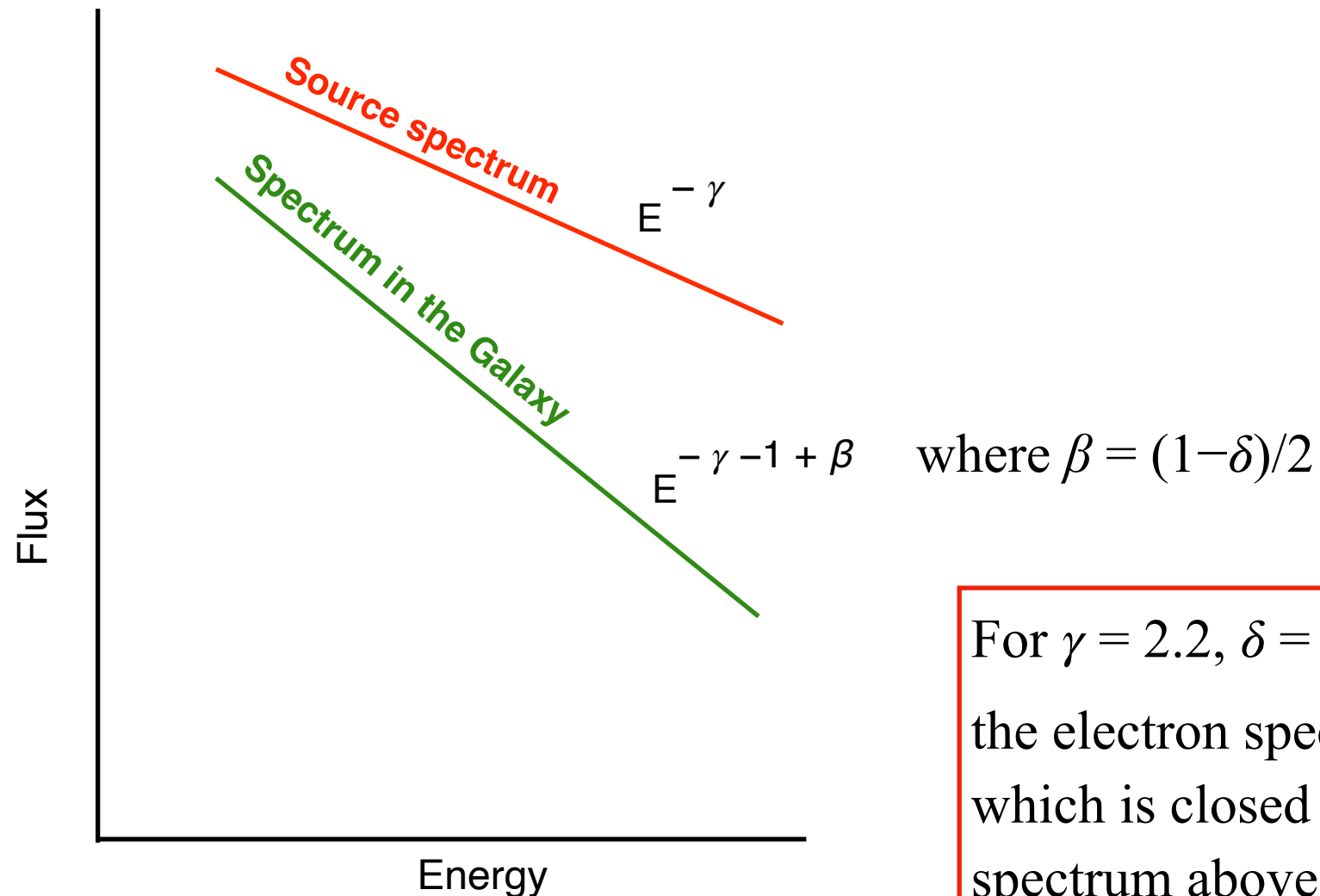
$$n_e(E) \propto \frac{1}{b(E)} \int_E^\infty dE' \frac{q(E')}{\sqrt{E^{\delta-1} - E'^{\delta-1}}} \quad \text{where, we take } b(E) = a E^2.$$

From Eq. (10), we see that for a power-law source spectrum such as $q(E) \propto E^{-\gamma_e}$, we get $n_e(E) \propto E^{-\gamma_e-1+\beta}$ where $\beta = (1 - \delta)/2$. Thus, the equilibrium electron spectrum in the Galaxy is modified mainly by the radiative energy losses with a small correction β arising due to the diffusive propagation effect.

Galactic propagation of CR electrons

The full transport equation in the Galaxy including diffusion and energy loss

CR electron spectrum from a stationary uniform source distribution:



For $\gamma = 2.2$, $\delta = 0.6$:
the electron spectrum follows E^{-3}
which is closed to the observed
spectrum above 10 GeV.

Galactic propagation of CR electrons

Time-dependent propagation equation:

$$\nabla \cdot (D \nabla N_e) + \frac{\partial}{\partial E} \{b(E) N_e\} + Q_e = \frac{\partial N_e}{\partial t}, \quad (18)$$

where $N_e(E, t)$ is the density of electrons with the kinetic energy E , $b(E) = -dE/dt$ is the energy-loss rate and $Q_e(\mathbf{r}, E, t)$ denotes the electron injection rate into the ISM.

**Note: This equation applies to a discrete (in space-time) single source in the Galaxy.
So, mainly to nearby sources.**

Galactic propagation of CR electrons

Time-dependent propagation equation:

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where $N_e(E, t)$ is the density of electrons with the kinetic energy E , $b(E) = -dE/dt$ is the energy-loss rate and $Q_e(\mathbf{r}, E, t)$ denotes the electron injection rate into the ISM. The Green function of equation (18) can be obtained as given below (see e.g. Ginzburg & Syrovatskii 1964; Gratton 1972)

$$G_e(\mathbf{r}, \mathbf{r}', E, E', t, t') = \frac{1}{8 [\pi f(E, E')]^{3/2} b(E)} \exp \left[\frac{-(\mathbf{r}' - \mathbf{r})^2}{4 f(E, E')} \right] \times \delta[t' - t + g(E, E')], \quad (19)$$

where

$$f(E, E') = \int_E^{E'} \frac{D(u)}{b(u)} du \quad \text{and} \quad g(E, E') = \int_E^{E'} \frac{1}{b(u)} du.$$

Galactic propagation of CR electrons

Time-dependent propagation equation:

For most studies, we can assume that the energy loss of the electrons is due to synchrotron and inverse Compton interactions which are true mostly for energies $\gtrsim 10$ GeV. We take

$$b(E) = aE^2, \quad (20)$$

where $a = 1.01 \times 10^{-16}(w_{\text{ph}} + w_B) \text{ GeV s}^{-1}$ and w_{ph} and w_B represent the energy densities in eV cm^{-3} for the background photons and the magnetic field, respectively. Equation (20) assumes that the inverse Compton scattering of the background photons occurs in the Thompson regime.

Galactic propagation of CR electrons

Time-dependent propagation equation:

The general solution of equation (18) is given by

$$N_e(\mathbf{r}, E, t) = \int_{-\infty}^{\infty} d\mathbf{r}' \int_E^{\infty} dE' \int_{-\infty}^t dt' G_e(\mathbf{r}, \mathbf{r}', E, E', t, t') \\ \times Q_e(\mathbf{r}', E, t'). \quad (21)$$

For an energy-independent burst-like injection of electrons at time t_0 , we take the source term as $Q_e(\mathbf{r}', E', t') = q(\mathbf{r}')q(E')\delta(t' - t_0)$, where $q(E') \propto E'^{-\Gamma}$ denotes the source spectrum.

Galactic propagation of CR electrons

Time-dependent propagation equation:

Now, setting $\mathbf{r} = 0$ and performing the integrals over E' and t' , equation (21) becomes

$$N_e(E, t) = \frac{q(E)}{8(\pi C)^{3/2}} \left(1 - \frac{E}{E_t}\right)^{\Gamma-2} \int_{\infty}^{-\infty} d\mathbf{r}' \quad \text{Burst-like injection} \\ \times \exp\left(\frac{-\mathbf{r}'^2}{4C}\right) q(\mathbf{r}'), \quad (22)$$

where $E_t = 1/(a(t - t_0))$ is the energy at which the energy-loss time is equal to $(t - t_0)$,

$$C = \frac{D(E)}{a(1 - \delta)E} \left[1 - \left(1 - \frac{E}{E_t}\right)^{1-\delta}\right] \quad (23)$$

and δ is the index of the diffusion coefficient. Equation (22) is valid for electrons with energies $E < E_t$. For $E > E_t$, $N_e = 0$.

Galactic propagation of CR electrons

Time-dependent propagation equation:

For a point source, the solution becomes:

Burst-like injection

$$N_e(E, t) = \frac{q_e(E)}{8(\pi C)^{3/2}} \left(1 - \frac{E}{E_t}\right)^{\Gamma-2} \exp\left(\frac{-r_s^2}{4C}\right), \quad (24)$$

where $E_t = 1/(a(t - t_0))$ and t_0 is the particle release time,

Also, $q_e(E) = k_e E^{-\Gamma}$ is the source spectrum for the point source.

From equation (23), we can see that in the energy region $E \ll E_t$ where the effect of the energy loss is less important, $C \rightarrow D(t - t_0)$ and equation (24) tends towards the point-source solution for CR protons (equation 7). Therefore, high-energy electrons whose $r_{\text{diff}} \gg r_s$ and $E \ll E_t$ have spectrum which follows $N_e(E) \propto E^{-(\Gamma + \frac{3}{2}\delta)}$, which is similar to the asymptotic solution of high-energy protons

Galactic propagation of CR electrons

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$$N_e(E, t) = \frac{q_e(E)}{8(\pi C)^{3/2}} \left(1 - \frac{E}{E_t}\right)^{\Gamma-2} \exp\left(\frac{-r_s^2}{4C}\right), \quad (24)$$

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- * The solution for “**continuous injection**” from a point source can be obtained by integration Eq. (24) over the injection time t_0 .
- * The solution will be similar to that of the “steady/stationary source” discussed before.

Galactic propagation of CR electrons

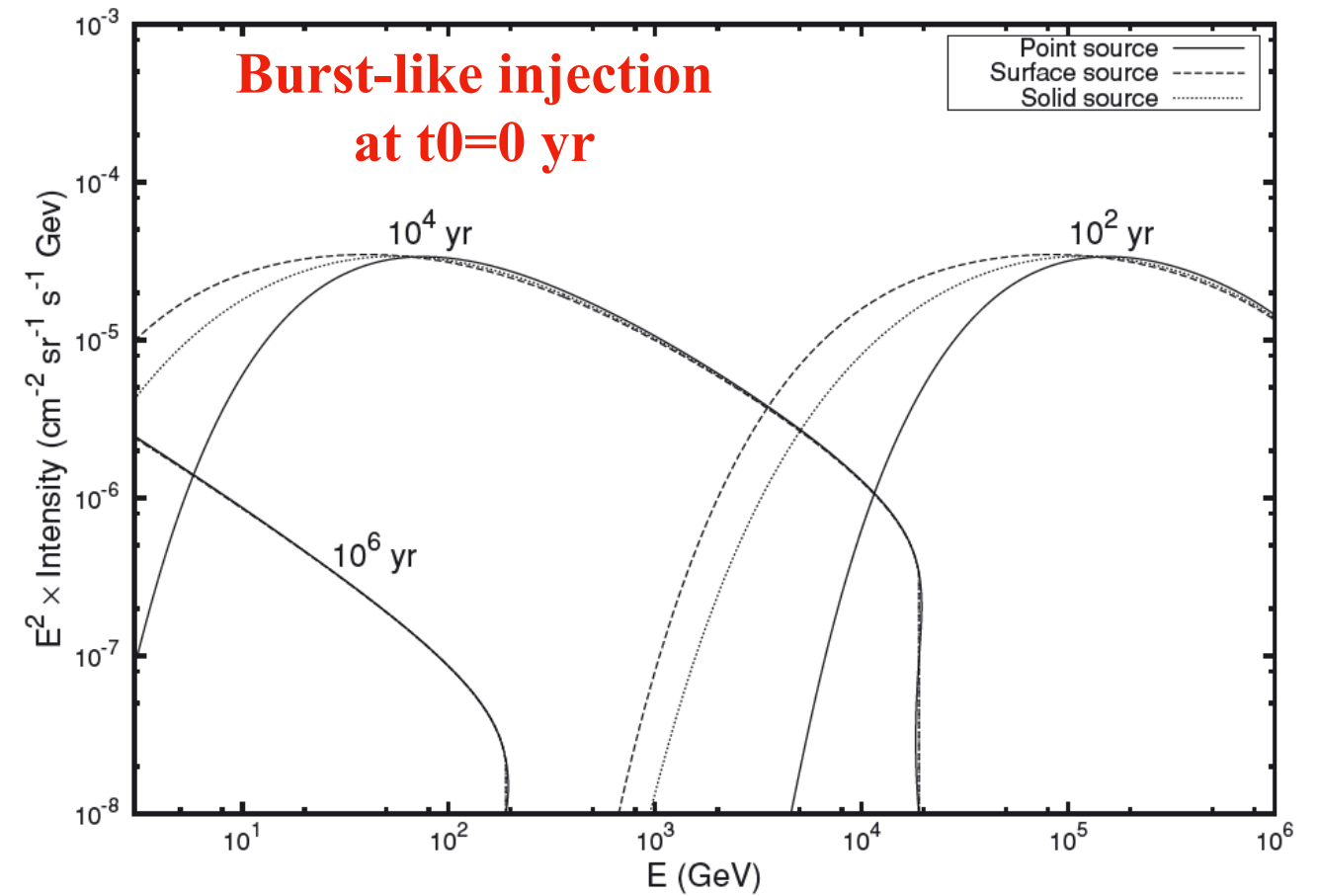
Time dependent CR electron spectrum from a point source

Particle release times: $t_0=0$ yrs. 

$$\Gamma = 2.0, \delta = 0.6, D_0 = 2.9 \times 10^{28} \text{ cm}^2 \text{ s}^{-1},$$
$$r_s = 0.2 \text{ kpc} \quad B = 6 \text{ uG}$$

$$w_{\text{MBR}} = 0.25 \text{ eV cm}^{-3}, w_{\text{op}} = 0.5 \text{ eV cm}^{-3}$$

Neglect the dashed and dotted lines.



Galactic propagation of CR electrons

Time dependent CR electron spectrum from a point source

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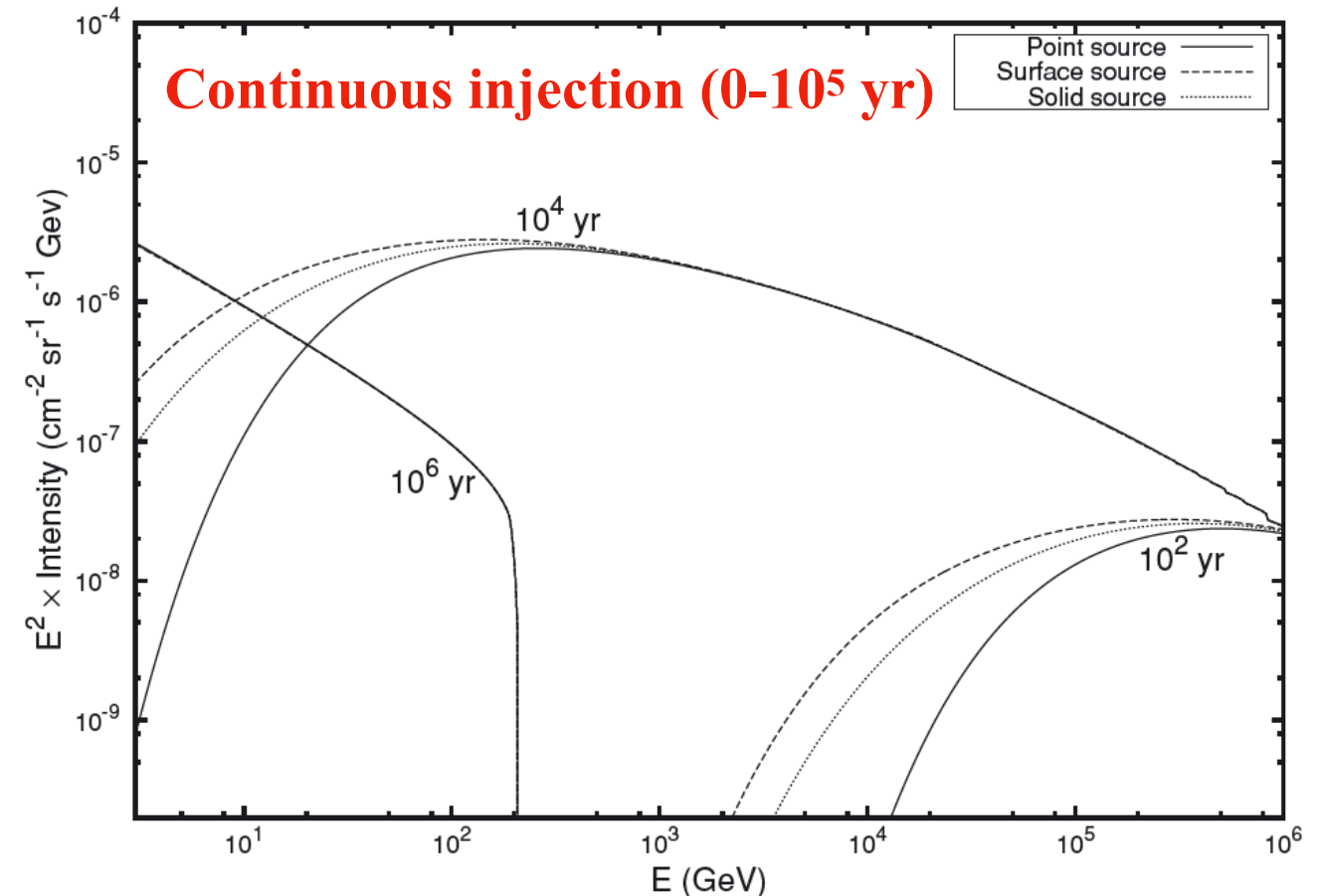
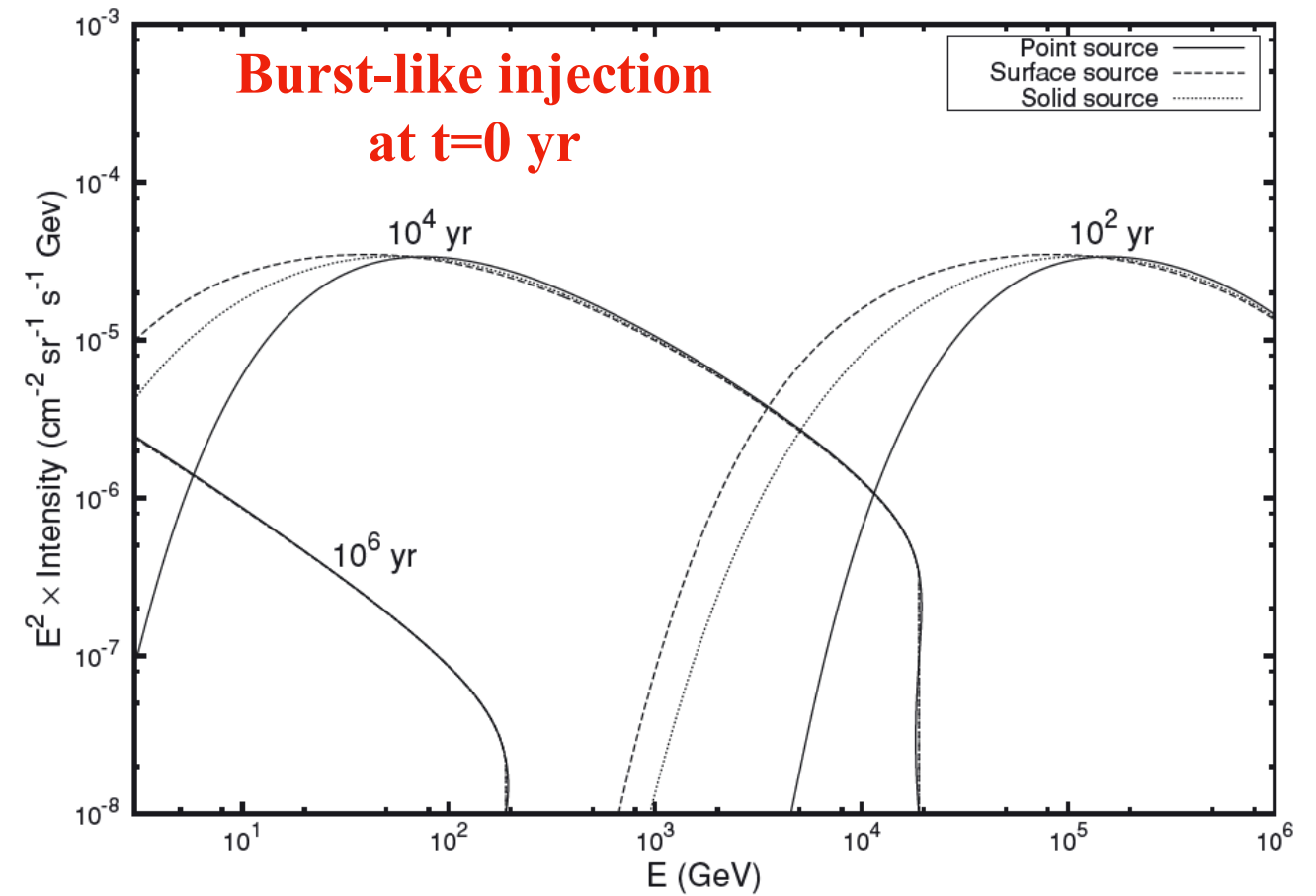
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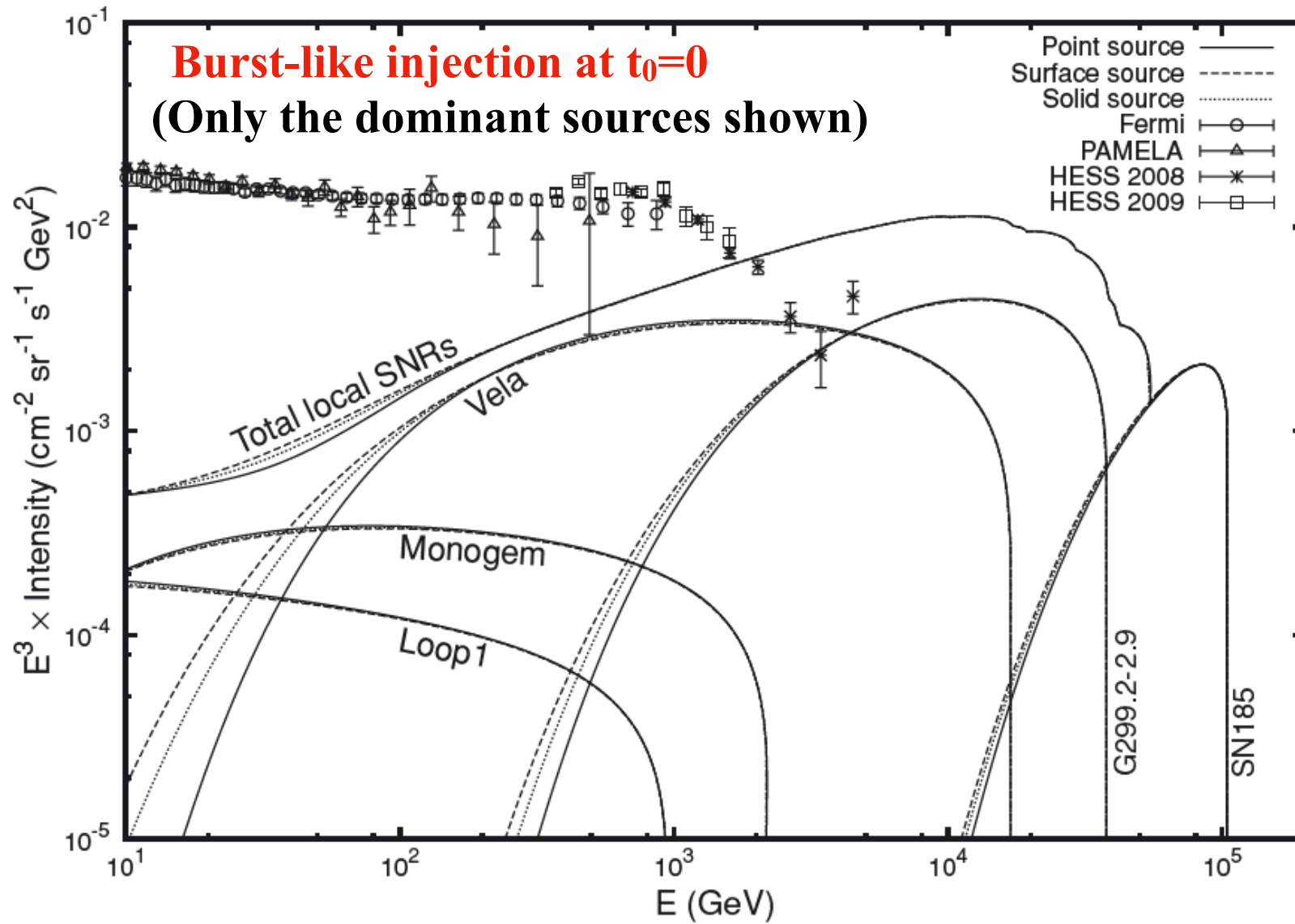
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Continuous injection for time: 0-10⁵ yr 



Galactic propagation of CR electrons

Application to the nearby supernova remnants (within 1 kpc from the Earth)



SNR	Distance (kpc)	Age (yr)
Cygnus Loop	0.540	1.0×10^4
HB21	0.800	1.9×10^4
HB9	0.800	6.6×10^3
S147	0.800	4.6×10^3
Vela	0.294	1.12×10^4
G299.2–2.9	0.500	5.0×10^3
SN185	0.950	1.8×10^3
Monogem	0.300	8.6×10^4
Geminga	0.157	3.4×10^5
Loop1	0.170	2.0×10^5
G114.3+0.3	0.700	4.1×10^4
Vela Junior	0.750	3.5×10^3

Neglect the dashed and dotted lines.

$$\Gamma = 2.3 ; \delta = 0.6, D_0 = 2.9 \times 10^{28} \text{ cm}^2 \text{ s}^{-1},$$

$$E_{\text{max}} = 10^5 \text{ GeV}$$

Note that the spectra will look different for different values of t_0 .

Galactic propagation of CR electrons

Application to the nearby supernova remnants (within 1 kpc from the Earth)

Most likely, high-energy electron suffer significant energy losses within the SNR itself before they are released into the ISM. Therefore, electrons at higher energies might be released with a spectrum steeper than the lower energy ones. So, it is reasonable to assume a high-energy cut-off at few TeVs in the source spectrum of electrons. Commonly used spectrum is an exponential cut-off in the form of $\exp(-E/E_c)$.

Galactic propagation of CR electrons

Application to the nearby supernova remnants (within 1 kpc from the Earth)

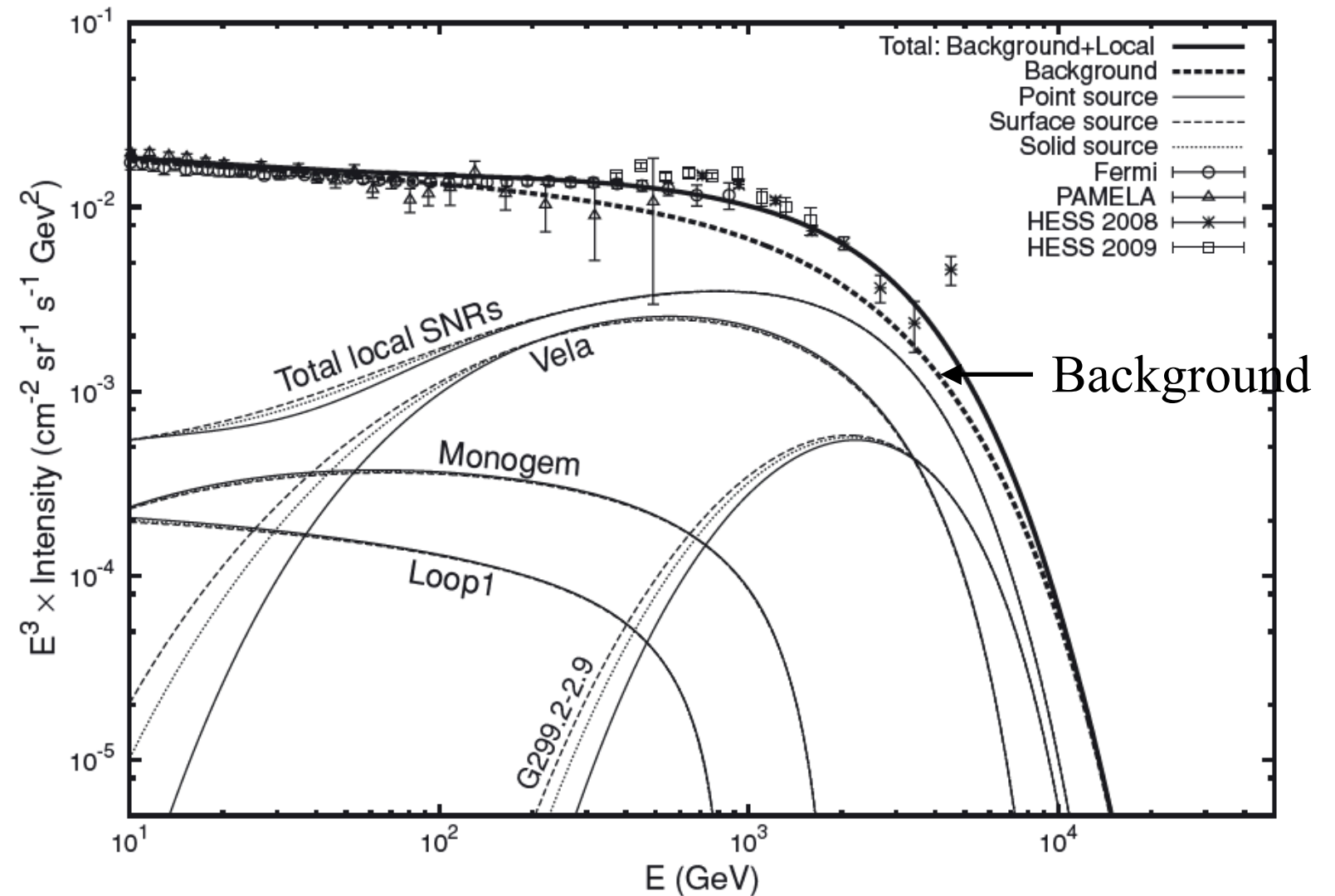
For an exponential cut-off in the source spectrum at $E_c=2$ TeV:

$$q_e(E) = k_e E^{-\Gamma} \exp(-E/E_c)$$

$$\Gamma = 2.3, \delta = 0.6,$$

$$D_0 = 2.9 \times 10^{28} \text{ cm}^2 \text{ s}^{-1},$$

Neglect the thin- dashed and the dotted lines.



Galactic propagation of CR electrons

Next, we will learn about the open issues with the standard model of CR origin based on new high-precision measurements.